

RuralConnect P2P - Project Stages

Stage 1: Registration & Onboarding

- Borrower uploads Aadhaar/ID documents (via DigiLocker or PDF).
- Provides biometric (face scan).
- System extracts and normalizes fields, generates hash, stores hash on blockchain.
- Profile created in MongoDB.
- Lender undergoes similar onboarding with Aadhaar/ID docs + OTP verification.
- Lender profile created in MongoDB.

Stage 2: Loan Request (Borrower)

- Borrower submits loan request (amount, purpose, repayment plan – monthly/seasonal).
- Optional collateral/asset documents uploaded, hash stored on blockchain.
- Request stored in backend database and forwarded to matching engine.

Stage 3: Loan Matching (System)

- Matching algorithm pairs borrower and lender.
- Factors: loan amount, interest, repayment ability, lender risk appetite.
- Borrower and lender both get notification of a proposed match.

Stage 4: Loan Agreement

- If lender accepts, a smart contract is created with loan terms.
- Both borrower and lender verify agreement via e-signature/OTP.
- Smart contract stored on blockchain to track lifecycle.

Stage 5: Fund Transfer

- Lender transfers loan via UPI/Payment Gateway to borrower's account.
- Transaction ID stored in backend and blockchain log.
- Borrower notified of funds received.

Stage 6: Repayment & Tracking

- Borrower repays via UPI/Bank, system updates repayment in database and blockchain.
- Notifications/reminders sent before due dates.
- Smart contract auto-updates repayment status.

Stage 7: Loan Closure / Default

- If full repayment is done, smart contract closes and blockchain marks loan as Closed.
- If repayment fails or defaults, smart contract flags it and borrower profile updated negatively.
- Lender notified of closure or default.